TASK 1: Basic Data Exploration

In [32]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

In [5]:
```
df = pd.read_excel("HousePrediction.xlsx")
print(df)
```

```
        Id  MSSubClass MSZoning  LotArea LotConfig BldgType  OverallCond  \
0        0          60       RL     8450    Inside     1Fam            5
1        1          20       RL     9600       FR2     1Fam            8
2        2          60       RL    11250    Inside     1Fam            5
3        3          70       RL     9550    Corner     1Fam            5
4        4          60       RL    14260       FR2     1Fam            5
...    ...         ...      ...      ...       ...      ...          ...
2914  2914         160       RM     1936    Inside    Twnhs            7
2915  2915         160       RM     1894    Inside   TwnhsE            5
2916  2916          20       RL    20000    Inside     1Fam            7
2917  2917          85       RL    10441    Inside     1Fam            5
2918  2918          60       RL     9627    Inside     1Fam            5

      YearBuilt  YearRemodAdd Exterior1st  BsmtFinSF2  TotalBsmtSF  SalePrice
0          2003          2003     VinylSd         0.0        856.0   208500.0
1          1976          1976     MetalSd         0.0       1262.0   181500.0
2          2001          2002     VinylSd         0.0        920.0   223500.0
3          1915          1970     Wd Sdng         0.0        756.0   140000.0
4          2000          2000     VinylSd         0.0       1145.0   250000.0
...         ...           ...         ...         ...          ...        ...
2914       1970          1970     CemntBd         0.0        546.0        NaN
2915       1970          1970     CemntBd         0.0        546.0        NaN
2916       1960          1996     VinylSd         0.0       1224.0        NaN
2917       1992          1992     HdBoard         0.0        912.0        NaN
2918       1993          1994     HdBoard         0.0        996.0        NaN

[2919 rows x 13 columns]
```

In [6]:
```
print(df.columns)
```

```
Index(['Id', 'MSSubClass', 'MSZoning', 'LotArea', 'LotConfig', 'BldgType',
       'OverallCond', 'YearBuilt', 'YearRemodAdd', 'Exterior1st', 'BsmtFinSF2',
       'TotalBsmtSF', 'SalePrice'],
      dtype='object')
```

In [7]:
```
df.head(100)
```

Out[7]:

	Id	MSSubClass	MSZoning	LotArea	LotConfig	BldgType	OverallCond	YearBuilt
0	0	60	RL	8450	Inside	1Fam	5	2003
1	1	20	RL	9600	FR2	1Fam	8	1976
2	2	60	RL	11250	Inside	1Fam	5	2001
3	3	70	RL	9550	Corner	1Fam	5	1915
4	4	60	RL	14260	FR2	1Fam	5	2000
...	...	...	...	...	...	...	...	...
95	95	60	RL	9765	Corner	1Fam	8	1993
96	96	20	RL	10264	Inside	1Fam	5	1999
97	97	20	RL	10921	Inside	1Fam	5	1965
98	98	30	RL	10625	Corner	1Fam	5	1920
99	99	20	RL	9320	Inside	1Fam	5	1959

100 rows × 13 columns

TASK 2: Data Cleaning & Analysis

Q1) Checking for Missing Values

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
Id                 0
MSSubClass         0
MSZoning           4
LotArea            0
LotConfig          0
BldgType           0
OverallCond        0
YearBuilt          0
YearRemodAdd       0
Exterior1st        1
BsmtFinSF2         1
TotalBsmtSF        1
SalePrice       1459
dtype: int64
```

Q2) Features with NaN Values

In [9]:
```
df.columns[df.isnull().any()]
```

Out[9]:
```
Index(['MSZoning', 'Exterior1st', 'BsmtFinSF2', 'TotalBsmtSF', 'SalePrice'], dt
ype='object')
```

Q3) Calculate Mean Sales Price for Missing/Present Information

In [10]:
```
df['BsmtFinSF2_missing'] = df['BsmtFinSF2'].isnull()
```

```
df.groupby('BsmtFinSF2_missing')['SalePrice'].mean()
```

Out[10]:

```
BsmtFinSF2_missing
False    180921.19589
True              NaN
Name: SalePrice, dtype: float64
```

Q4) Count of Numerical Features

In [11]:

```
numerical_features = df.select_dtypes(include=['int64','float64'])
print(len(numerical_features.columns))
```

```
9
```

Q5) Print the First Five Rows of Numerical Values

In [12]:

```
numerical_features.head()
```

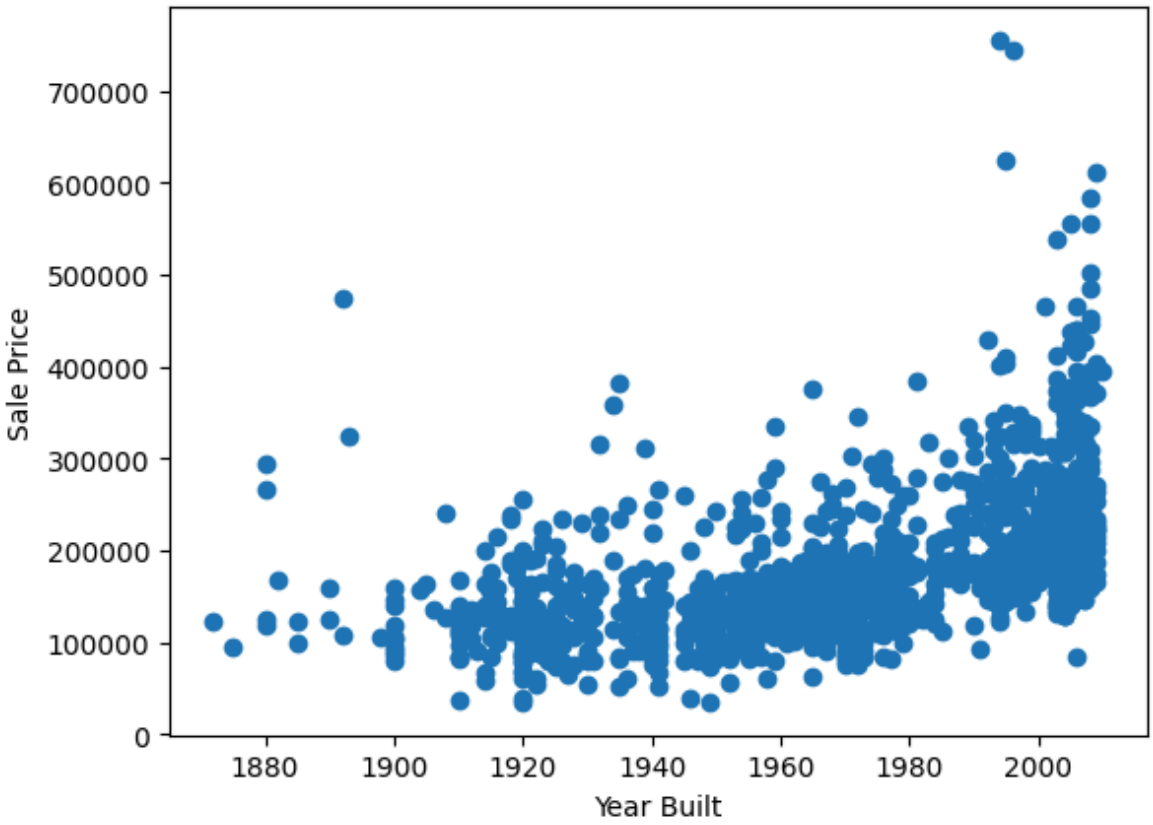
Out[12]:

	Id	MSSubClass	LotArea	OverallCond	YearBuilt	YearRemodAdd	BsmtFinSF2	Total
0	0	60	8450	5	2003	2003	0.0	
1	1	20	9600	8	1976	1976	0.0	
2	2	60	11250	5	2001	2002	0.0	
3	3	70	9550	5	1915	1970	0.0	
4	4	60	14260	5	2000	2000	0.0	

Q6) Compare the Difference Between Year Features and SalePrice

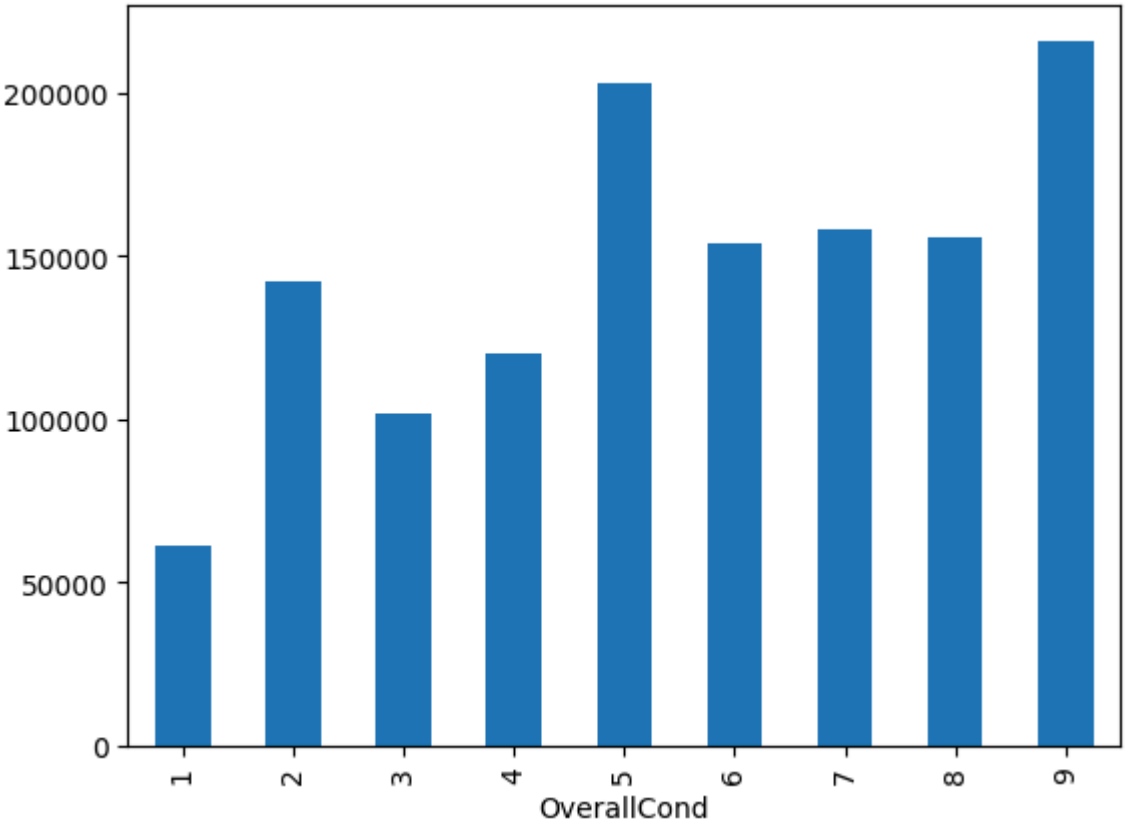
In [13]:

```
plt.scatter(df['YearBuilt'], df['SalePrice'])
plt.xlabel("Year Built")
plt.ylabel("Sale Price")
plt.show()
```



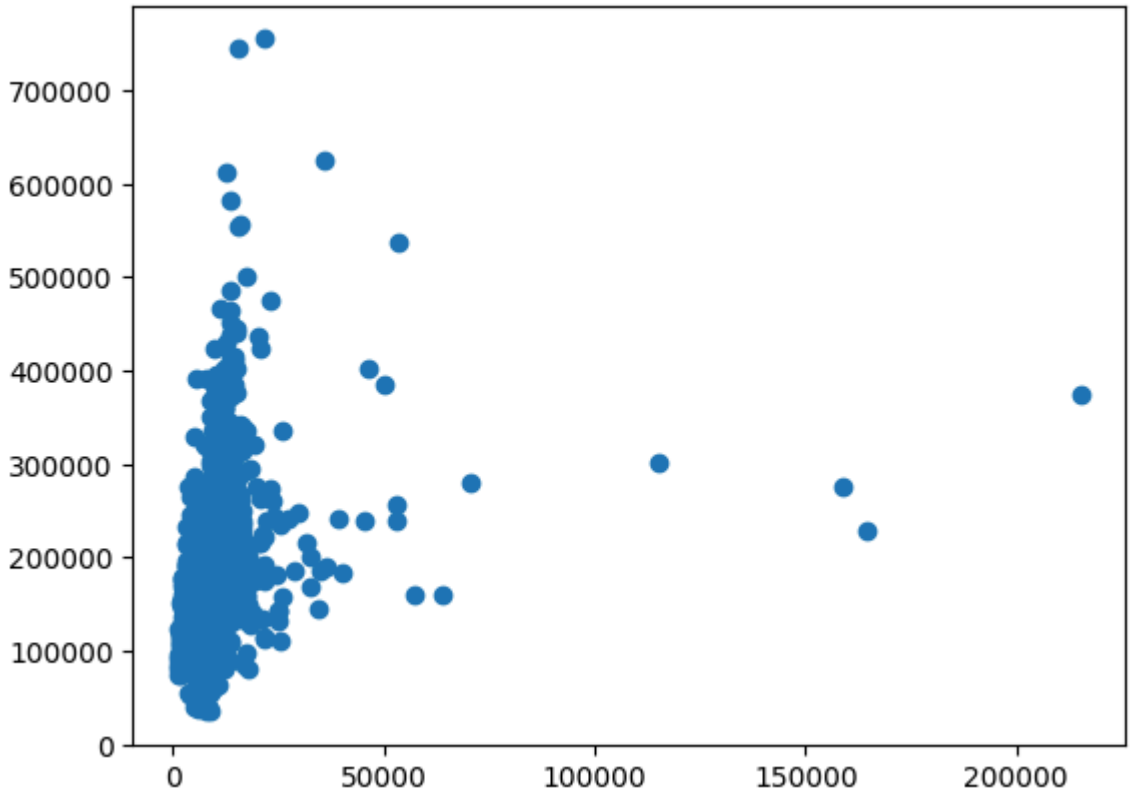
Q7) Relationship Between Discrete Variables and Sales Price

In [14]:
```
df.groupby('OverallCond')['SalePrice'].mean().plot(kind='bar')
plt.show()
```



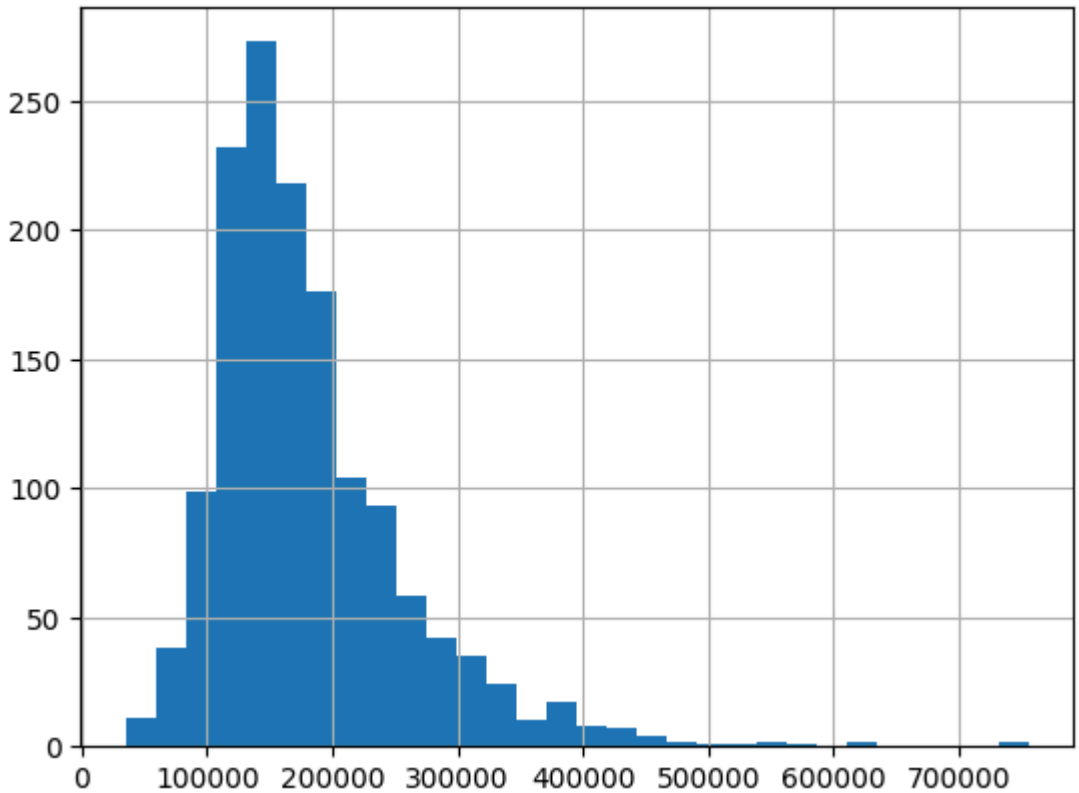
Q8) Relationship Between Continuous Variables and Sales Price

In [15]:
```
plt.scatter(df['LotArea'], df['SalePrice'])
plt.show()
```



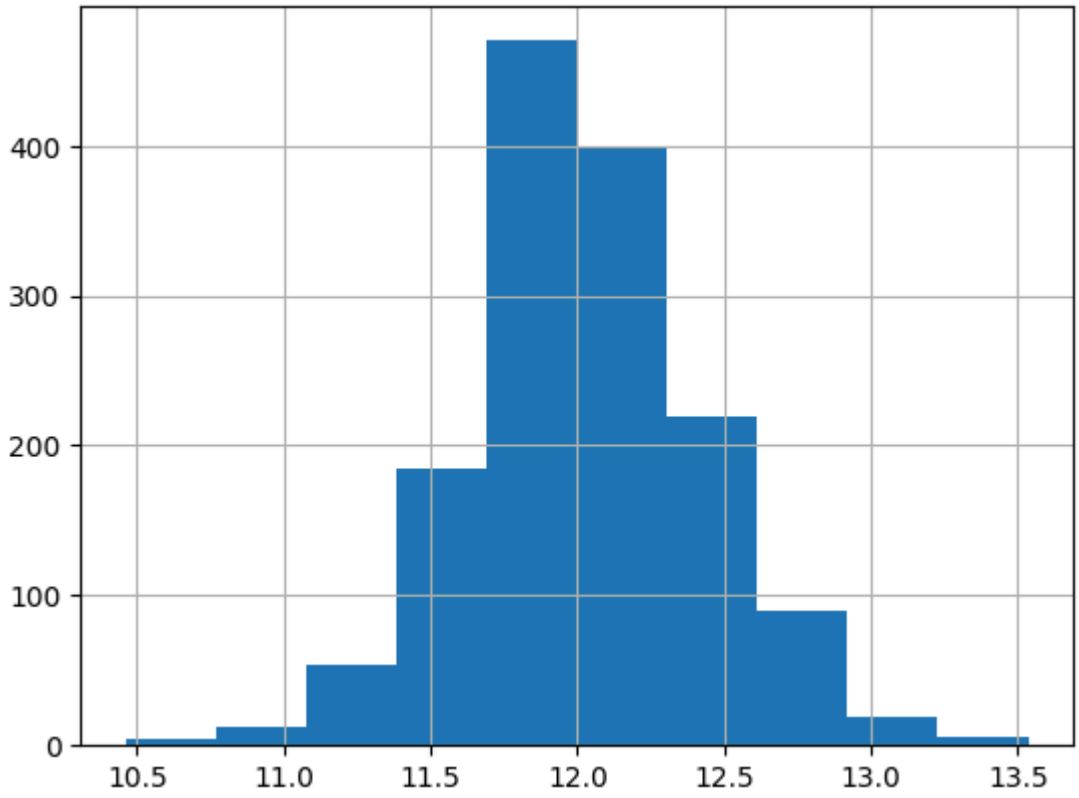
Q9) Histogram Analysis for Continuous Variables

In [17]:
```
df['SalePrice'].hist(bins=30)
plt.show()
```



Q10) Logarithmic Transformation

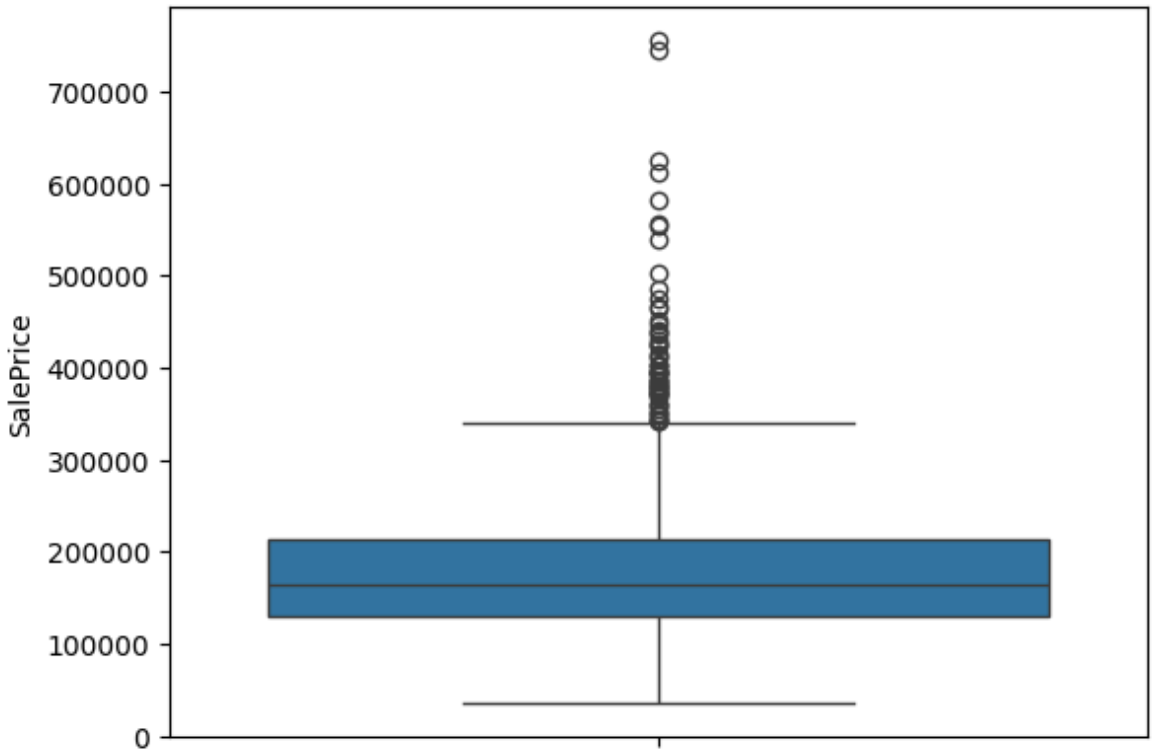
In [18]:
```
df['LogSalePrice'] = np.log(df['SalePrice'])
df['LogSalePrice'].hist()
plt.show()
```



Task 3: Feature Engineering and Advanced Analysis

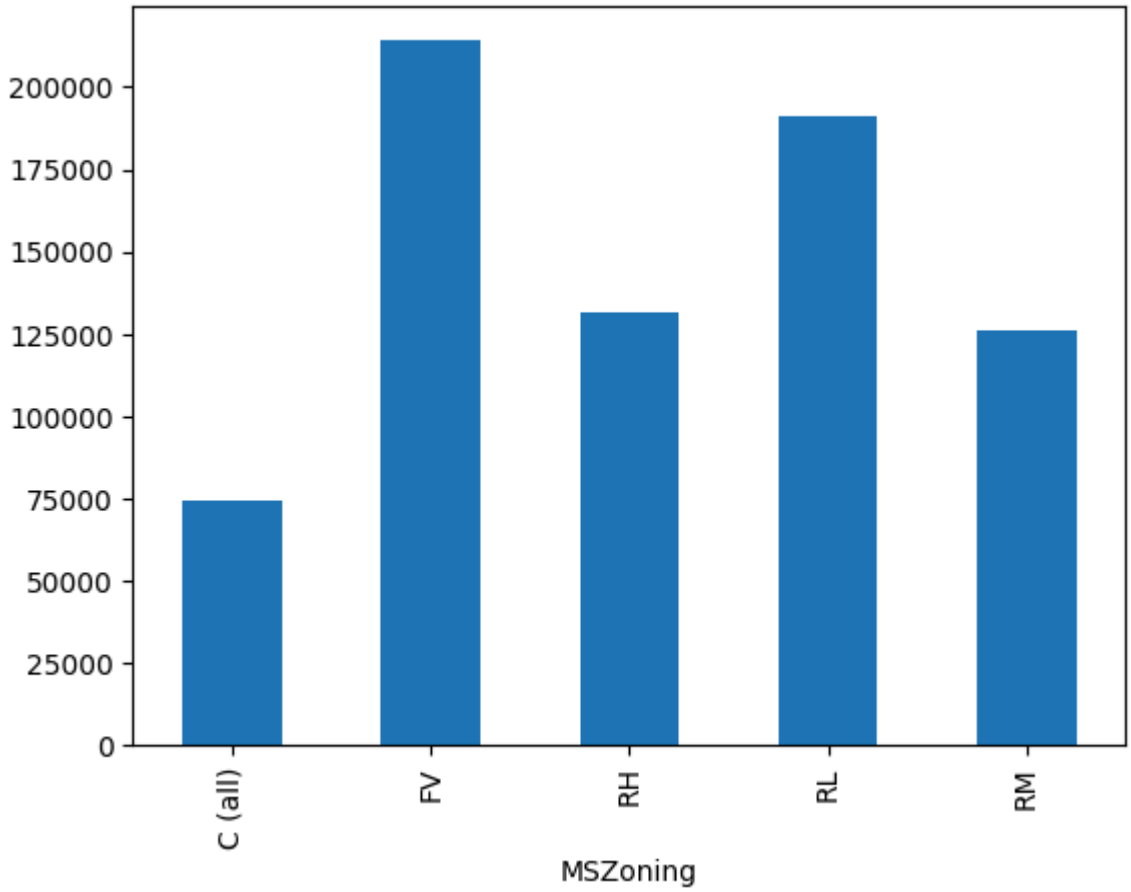
Q1) Find Outliers

In [19]:
```
sns.boxplot(df['SalePrice'])
plt.show()
```



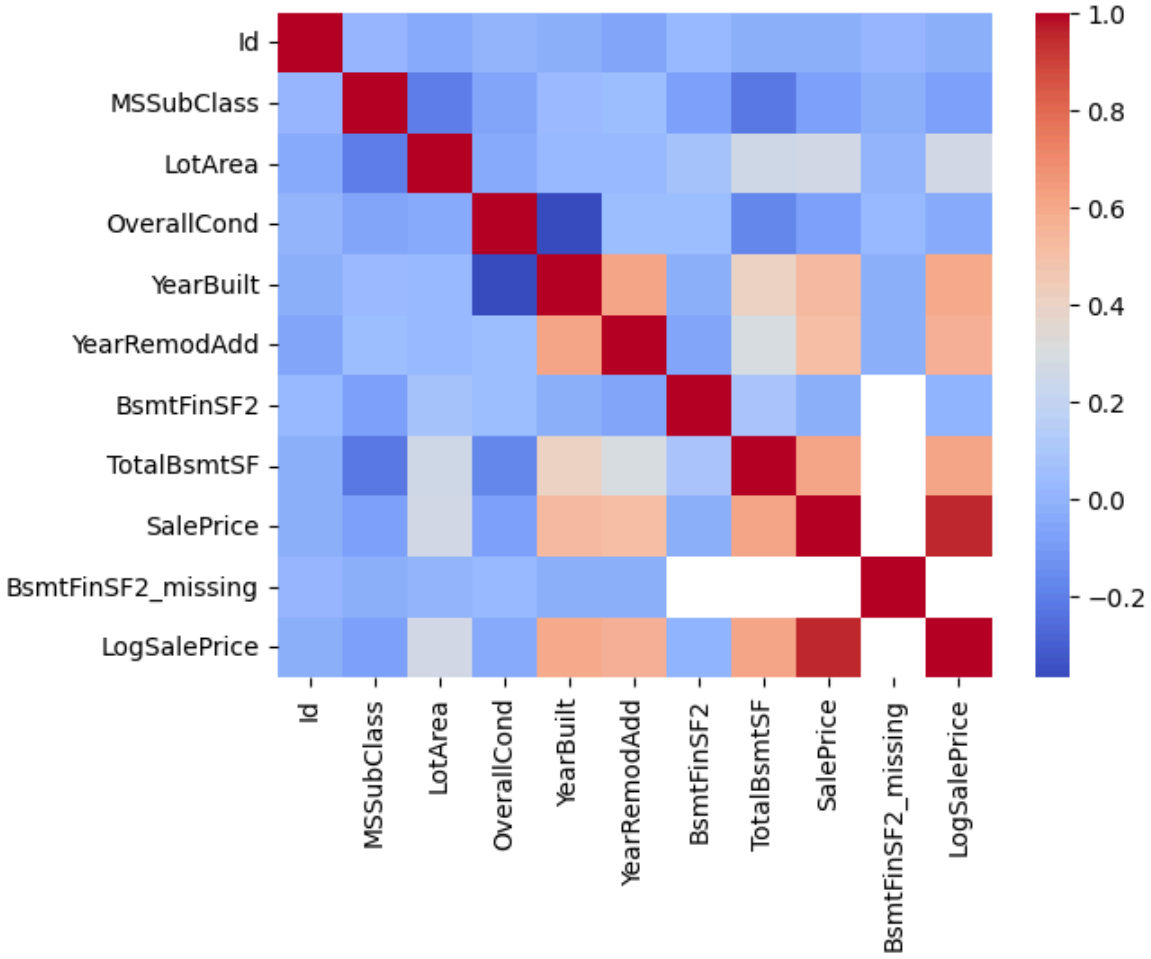
Q2) Relationship Between Categorical Features and SalePrice

In [20]:
```
df.groupby('MSZoning')['SalePrice'].mean().plot(kind='bar')
plt.show()
```



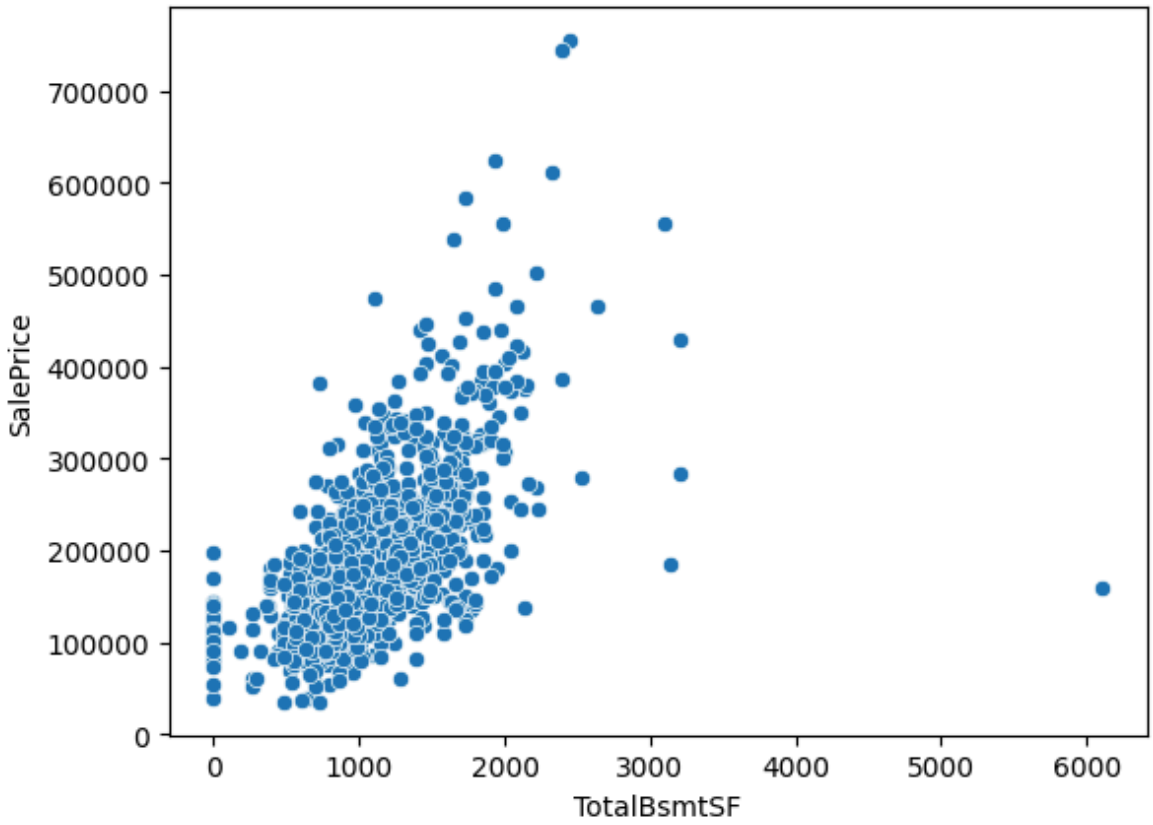
3) Correlation Between Numerical Features and SalePrice

In [21]:
```
corr = df.corr(numeric_only=True)
sns.heatmap(corr, cmap='coolwarm')
plt.show()
```



Q4) Continuous Features vs. SalePrice

In [22]:
```
sns.scatterplot(x=df['TotalBsmtSF'], y=df['SalePrice'])
plt.show()
```



Q5) Feature Engineering

Handle Missing Values:

In [27]:
```
df.fillna(df.median(numeric_only=True), inplace=True)
print(df)
```

```
        Id  MSSubClass  LotArea  OverallCond  YearBuilt  YearRemodAdd  \
0        0          60     8450            5       2003          2003
1        1          20     9600            8       1976          1976
2        2          60    11250            5       2001          2002
3        3          70     9550            5       1915          1970
4        4          60    14260            5       2000          2000
...    ...         ...      ...          ...        ...           ...
2914  2914         160     1936            7       1970          1970
2915  2915         160     1894            5       1970          1970
2916  2916          20    20000            7       1960          1996
2917  2917          85    10441            5       1992          1992
2918  2918          60     9627            5       1993          1994

      BsmtFinSF2  TotalBsmtSF  SalePrice  BsmtFinSF2_missing  ...  \
0            0.0        856.0   208500.0               False  ...
1            0.0       1262.0   181500.0               False  ...
2            0.0        920.0   223500.0               False  ...
3            0.0        756.0   140000.0               False  ...
4            0.0       1145.0   250000.0               False  ...
...          ...          ...       ...                 ...  ...
2914         0.0        546.0   163000.0               False  ...
2915         0.0        546.0   163000.0               False  ...
2916         0.0       1224.0   163000.0               False  ...
2917         0.0        912.0   163000.0               False  ...
2918         0.0        996.0   163000.0               False  ...

      Exterior1st_CemntBd  Exterior1st_HdBoard  Exterior1st_ImStucc  \
0                   False                False                False
1                   False                False                False
2                   False                False                False
3                   False                False                False
4                   False                False                False
...                   ...                  ...                  ...
2914                 True                False                False
2915                 True                False                False
2916                False                False                False
2917                False                 True                False
2918                False                 True                False

      Exterior1st_MetalSd  Exterior1st_Plywood  Exterior1st_Stone  \
0                   False                False              False
1                    True                False              False
2                   False                False              False
3                   False                False              False
4                   False                False              False
...                   ...                  ...                ...
2914                False                False              False
2915                False                False              False
2916                False                False              False
2917                False                False              False
2918                False                False              False

      Exterior1st_Stucco  Exterior1st_VinylSd  Exterior1st_Wd Sdng  \
0                  False                 True                False
1                  False                False                False
2                  False                 True                False
3                  False                False                 True
4                  False                 True                False
...                  ...                  ...                  ...
2914               False                False                False
```

```
2915                False                False                False
2916                False                 True                False
2917                False                False                False
2918                False                False                False

      Exterior1st_WdShing
0                   False
1                   False
2                   False
3                   False
4                   False
...                   ...
2914                False
2915                False
2916                False
2917                False
2918                False

[2919 rows x 37 columns]
```

Handle Categorical Variables:

In [36]:
```
le = LabelEncoder()
for col in df.select_dtypes(include='object'):
    df[col] = le.fit_transform(df[col])
```

Handling Temporal Variables:

In [34]:
```
df['HouseAge'] = 2025 - df['YearBuilt']
print(df['HouseAge'])
```

```
0        22
1        49
2        24
3       110
4        25
       ...
2914     55
2915     55
2916     65
2917     33
2918     32
Name: HouseAge, Length: 2919, dtype: int64
```

In [ ]: